

Wearable Smart Insole for Early Diabetic Foot Ulcer Prediction Using Multi-Modal Sensing

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Abstract— *Diabetic foot ulcers (DFUs), a chronic complication of diabetes, accounts for severe infections, more than 85% of diabetes-related amputations, and increased mortality rate, if not detected and managed at an early stage. Continuous foot health monitoring of diabetes patients is critical to prevent severe outcomes. In this paper, a smart insole system is designed for continuous monitoring of vital foot health parameters via integrated multi-sensor technology. Six pressure sensors FSR 402, two temperature and humidity sensors SHT31, and one pulse oximeter sensor MAX30102, are mounted on the insole in accordance with the clinically relevant, high-risk areas. Data from these sensors is wirelessly communicated to the monitoring graphical user interface (GUI), offering real-time visualization and alerts. Performance of this insole was then evaluated on a set of individuals with different BMI ranges, in different postures and positions, and a versatile database was collected. A Long Short-Term Memory (LSTM) model is trained on the acquired dataset, which then predicts distinct pressure changes on six DFU prone areas, localized temperature variations, and peripheral blood oxygen levels of individuals and alerts them about the possible likelihood of an ulcer. This validates the insole's utility as an early proactive monitoring device for DFU detection and prevention.*

Index Terms—Diabetic foot ulcer, smart insole, wearable sensors, plantar pressure monitoring, real-time health monitoring, AI, Deep learning.

I. INTRODUCTION

Diabetes is one of the leading diseases worldwide, currently affecting approximately 537 million people, with projections reaching 783 million by 2045 [1]. Diabetic foot ulcer (DFU) is a prevalent condition in individuals with diabetes mellitus, with a reported 5-year mortality rate of approximately 50%, highlighting its significant impact on patient survival. Individuals with diabetes who suffer from this complication are at an increased risk for lower limb amputations[2]. Early detection and management can

significantly reduce the risk of DFU. Major risk factors for DFUs include peripheral neuropathy and poor blood circulation [3]. Unfortunately, traditional clinical foot exams are intermittent and largely subjective, so ulcers can go undetected until advanced. Thus, there is a clear need for continuous, quantitative foot monitoring to catch early warning signs for the early identification of diabetic foot ulcers [4]. Studies show that excessive plantar pressure and shear stress are critical contributors to ulcer formation and serve as early indicators of DFU [5]. Excessive plantar pressure induces recurrent stress on particular areas of the foot, directly resulting in tissue degradation [6][7][8]. Furthermore, temperature can be another symptom to consider, as increased skin temperature frequently indicates initial inflammation, a recognized precursor to ulcer development. Moreover, irregular moisture levels, whether due to excessive perspiration or skin desiccation can compromise the skin's natural barrier, infections, and ulcers [9]. Collectively, these factors offer an extensive overview of foot health before open wounds develop. In this regard, various smart insoles embedded with different types of pressure sensors have been proposed over the past few years, for continuous monitoring of plantar pressure to detect DFU and identify high-risk areas [10]. Another study revealed that monitoring pressure alone is not enough, as pressure may vary with age and due to other complications as well, therefore, multi-modal sensing must be done for accurate analysis of DFU risks [11].

In this work, we build on these insights by integrating multiple low-cost sensors into an IoT-enabled insole, designed to monitor key parameters in real time and alert users or clinicians to potential DFU risks. It measures plantar pressure, temperature, and moisture level on clinically relevant high-risk areas, providing real-time visualizations and alerts. A versatile database is collected by testing the performance of the developed insole on a group of volunteers, during daily activities. A sequential deep learning model i.e., LSTM is trained on the acquired dataset to consider the pressure changes at six different points, localized temperature variations and moisture

changes and predict the possibility of ulcer formation accordingly achieving 95% accuracy.

II. METHODOLOGY

The proposed methodology involves insole development, data acquisition via microcontroller, performance evaluation and dataset acquisition, followed by AI framework training and GUI development. Overall schematic of the proposed smart system for continuous monitoring of foot health parameters is depicted in **figure1**.

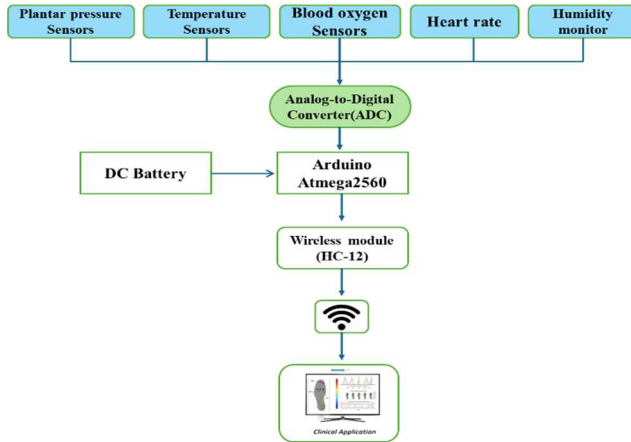


Figure 1: Schematic of the proposed methodology.

A. Insole development

The goal of the design is to make a wearable device that is comfortable and easy to use and can measure a number of biomarkers, including plantar pressure, temperature, humidity, heart rate and blood oxygen level. The smart insole in accordance with anatomical accuracy and comfortability; the flexible silicone and textile layers house the sensors without compromising user mobility. Six Pressure Sensors (FSR 402) were embedded in key plantar regions: two under the heel, three across metatarsal heads, and one under the hallux (toe); these sensors detect localized pressure changes associated with areas prone to ulceration. Two SHT31 digital temperature and humidity sensors were placed to measure skin temperatures at the heel and forefoot, while a MAX30102 pulse oximeter sensor was located adjacent to the hallux to record blood oxygen saturation (SpO₂) and heart rate which are shown in the **Figure 2**.

B. Electronics and Data Acquisition:

An Arduino art mega 2560 microcontroller is employed due to its low power consumption and built-in transmission capabilities, configured to read analog signals from FSR-402 sensors and digital I²C data from SHT31 and MAX30102 sensors, processed locally,

transmitted via HC-12 module to a custom-developed insole. An analog to-digital converter (ADC) sends the sensor values to the microcontroller like UART Communication. A 9V rechargeable lithium-ion battery served as the power supply with a battery life of approximately 7 hours.

C. Participant Study and Data Collection:

The system was evaluated on a group of volunteers ranging from ages 15 to 30. Every participant wore the insole while engaging in a set of predetermined exercises, including resting, walking, standing, sitting, toe raising, and stair climbing. For offline analysis, data from every sensor was continuously gathered and stored.

D. Data Analysis

Preprocessing was done on the sensor readings to eliminate noise and anomalies. A reference load cell was used to calibrate and normalize the pressure data from the FSR 402 so that they could be compared across subjects. To find indications of inflammation, temperature differences between foot regions were calculated. Trends in SpO₂ were tracked in order to identify any notable decreases during exercise. Sensor outputs from various activity conditions and sensor locations were analyzed using descriptive statistics and paired comparisons. In order to apply AI, time-series sensor data was analyzed using LSTM model, which allowed for the identification of temporal patterns associated with activity levels and physiological changes. Three categories (**no ulcer**, **mild infection**, and **ulcer**) were used to train the LSTM model to diagnose foot conditions. Temperature fluctuations, SpO₂ levels, and activity data gathered from wearable sensors were among the input features. Python and Tensor Flow were used for model implementation. It was trained using a labeled dataset of foot health data that has undergone the proper standardization and preprocessing. On the validation set, the trained LSTM model demonstrated its efficacy in assisting with the early detection and monitoring of foot problems, achieving an accuracy of around (90-95%).

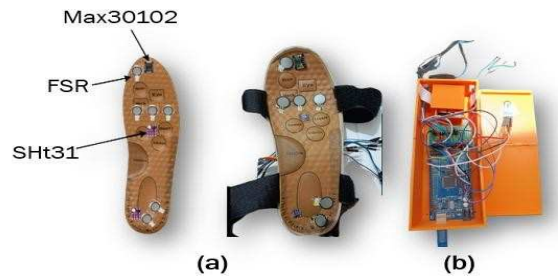


Figure 2: (a) Prototype Design (b) Assemble Prototype for Data Acquisition.

III. EXPERIMENTAL SETUP

The device's initial experimental objective was to develop a working smart Insole system design that could identify variations in blood oxygen levels, plantar pressure, humidity, heart rate and temperature at the appropriate points on the foot. Assuring that each sensor can identify values at its designated locations demonstrates that any deviations from the healthy baseline will be picked up at the appropriate spot on the foot where the sensor is located in **Figure 3**. This area can be brought up considered a matter of concern and addressed appropriately if it is verified that each sensor can accurately read and output values at its specified place on the foot. The system seeks to evaluate the sensitivity and ability of each sensor to detect variations in temperature, heart rate, humidity, blood oxygen levels and plantar pressure in their specific foot regions.

There were 22 people in the study, 6 women and 16 men out of which 5 were diabetes patient. They were all between the ages of 15 and 30. The volunteers were chosen to reflect diverse body types and shoe sizes. The age, weight, height, and shoe size of each participant were documented. The Smart insole device and guidelines for its correct insertion into the participant's insole were provided to them. Subsequently, the testing phase commenced. People who took part were mandated to wear the insole system for given time 5 to 10 minutes during the data collection session. The research team observed and assisted in the insertion of the insole into the participant's right shoe. To ensure proper functionality and data precision, all implanted sensors comprising temperature, plantar pressure, SpO₂, humidity, and heart rate were activated and assessed. The system utilized an HC-12 wireless module via UART communication for data transfer, facilitating a reliable long-range connection between a receiver linked to the computer-based data collecting system and a transmitter integrated with the insole. This guaranteed a steady and dependable connection during the tests, eliminating the necessity for a wireless communication. After sensor verification, participants received instructions to do a series of tasks that simulated real-world applications. The tasks encompassed both standard walking and rapid foot motions, including walking or mild jogging. The sensors collected physiological data in real time during these actions and observed dynamic variations in pressure, temperature, SpO₂, heart rate, and humidity

across different foot regions during movement by analyzing transmitted data through a MATLAB software interface that mapped and showed real-time sensor values.

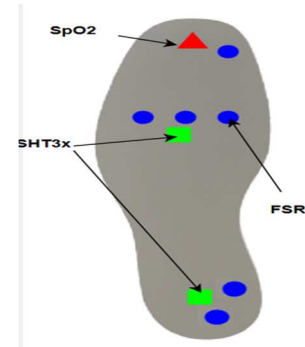


Figure 3: Sensor Value Mapping.

IV. RESULTS & DISCUSSION

The **Figure 4** shows the complete assembling of smart insole for monitoring diabetic foot step by step, starting with putting the sensors in place and making sure they fit securely. It starts with putting together the electronics and the parts, and ends with the completed wearable system, which includes strap fittings and a leg-mounted enclosure.

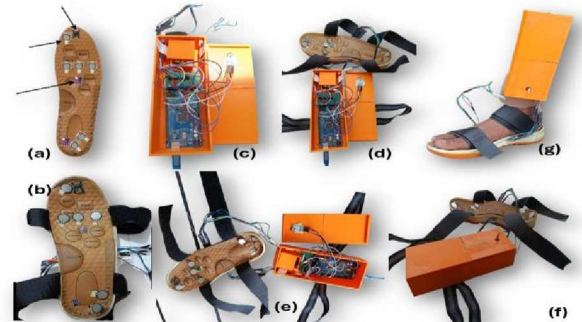


Figure 4: Complete Insole development Cycle.

In **Figure 5**, the right sided graph shows FSR sensor values over time show peak forces (red dots) per gait cycle, with dashed lines connecting peak load transitions. On the left, sensor values per cycle reflect overall sensitivity. Around cycle 45, a sharp surge suggests possible calibration issues or an anomaly. Other cycles remain stable, representing typical gait pressure. **Figure 6** reflects SHT31 sensor's recorded temperature data over different segments of time in seconds.

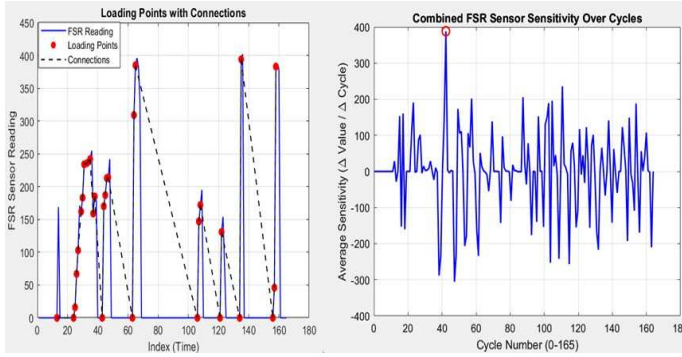


Figure 5: FSR loading points and Average Sensitivity over time.

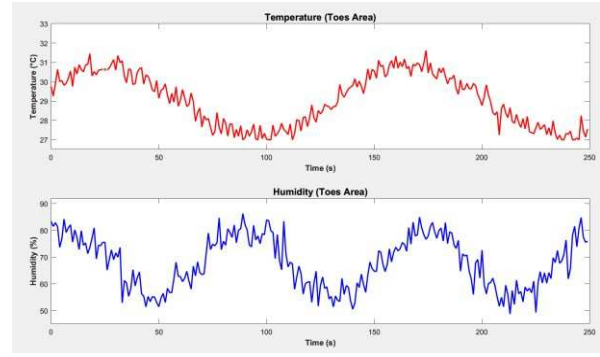


Figure 6: The SHT31 sensor recorded temperature data from the foot over time.

4.1. Participants' General Characteristics

There were 22 people in the trial (6 men and 16 women), and their anthropometric data is presented in **Table 1** below.

Age Group	Heel FSR	M1 FSR	M2 FSR	M3 FSR	Toes1 FSR	Toes2 FSR	SHT31 Temp (°C)	SHT31 Humidity (%)	BMI	Height (m)
12-15	689.7	802.7	847.3	805.8	779.2	815.5	33.1	48.7	22.4	1.6
15-18	861.7	900.4	889.5	923.7	764.1	777.3	32.6	47.4	23	1.65
18-19	891.25	291.47	355.64	306.99	661.64	655.67	32.69	47.16	21.78	1.5
19-21	826.3	921.5	712.7	777.2	794.4	830.3	32.3	46.9	22.1	1.67
21-22	885.4	463.36	514.72	479.72	693.02	696.46	32.72	47.38	21.99	1.535
22-25	882	735.3	936.5	630.4	873.7	933.8	33.3	49.6	23.4	1.71
26	879.55	635.25	673.8	652.45	724.4	737.25	32.75	47.6	22.2	1.57

Table 1: General Characteristics of Participants (N = 22)

4.2. Results Analysis

We looked at temperature data from twenty people doing six different things at four different foot positions. **Table 2** displays the average temperature for each activity, taking into account all locations and participants.

Activity	Temperature (°C)	Humidity (%)	SpO ₂ (%)
Fast Walking	28 - 30	60 - 68	92 - 95
Sitting with Feet on the Ground	30 - 32	57 - 65	89 - 93
Sitting with Feet Up	27 - 29	55 - 59	83 - 97
Standing	28 - 30	63 - 67	75 - 94
Walking	30 - 33	67 - 75	75 - 95

Table 2: Average temperature range for each sensor foot region.

4.3. Gait Phase Analysis

Figure 7(a) displays the complete Stick diagram of the complete Gait Phase cycle showing the potential pressure points during a complete Gait phase. It covers all possible pressure points during a Gait cycle starting from Heel, Foot, Mild stance and Toe area. **Figure 7(b)** shows the Heat maps of all sensors during each phase of Gait phase which helps analyzing which sensor is under pressure at a specific foot position.

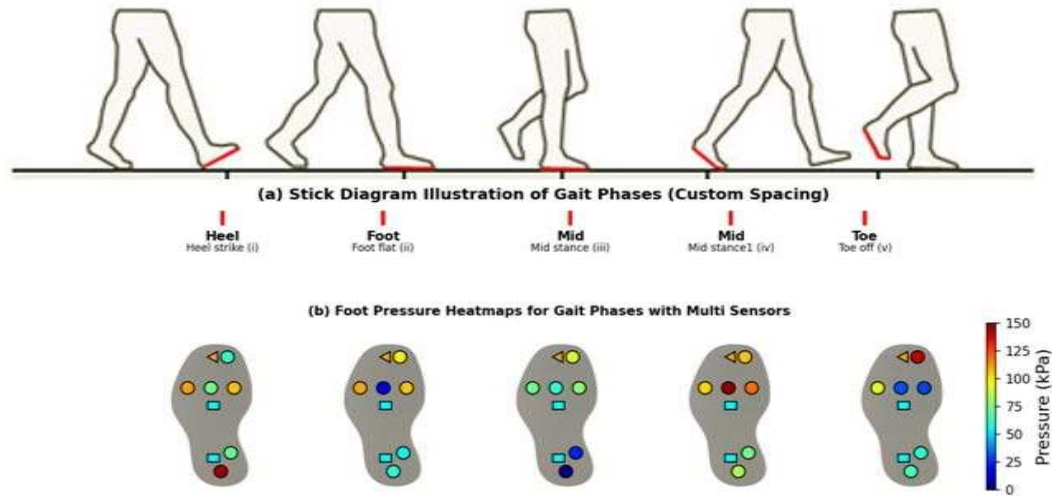


Figure 7: Illustration of pressure mapping by the sensors for various foot positions.

4.4. AI Implementation Results Analysis

A machine learning model utilizing Long Short Term Memory (LSTM) networks was created to classify high-risk foot diseases based on time-series sensor data. The input features comprised pressure patterns, temperature gradients, and SpO₂ fluctuations recorded throughout diverse physical activities. The LSTM model attained an accuracy of 94 - 96% in identifying aberrant pressure zones and early indicators of inflammation. These findings underscore the efficacy of AI-driven analytics in delivering prompt notifications for the prevention of diabetic foot ulcers. The Confusion matrix, training Vs validation accuracy and training Vs Validation loss can be seen below. **Figure 8** shows the model's performance evaluation results in terms of loss, accuracy and confusion matrix.

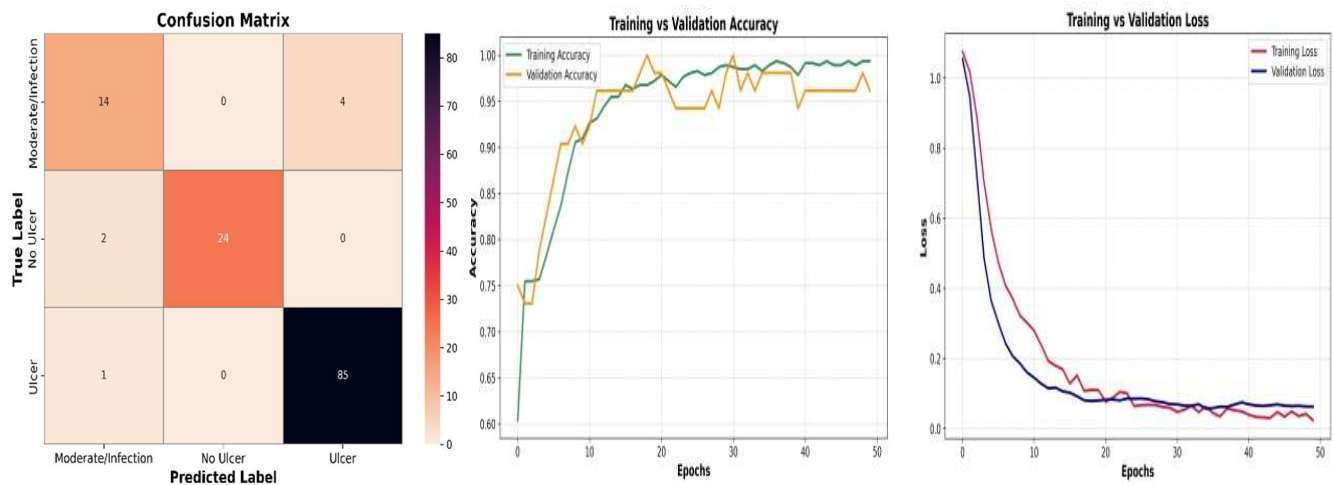


Figure 8: Confusion Matrix, training vs validation accuracy and training vs validation loss of proposed model

V. CONCLUSION

We have developed a smart wearable device that tracks plantar pressure, skin temperature, humidity, heart rate and oxygenation to identify early diabetic foot ulcer risk. By integrating various modalities, the device provides a more comprehensive assessment than any individual measurement alone. Simultaneously detecting a pressure hotspot and localized heating offers more compelling evidence of tissue stress.

The prototype performance suggests its potential usefulness for diabetic foot care. In the future, data from this device may be utilized in machine learning algorithms for predictive analytics. Recent studies have employed machine learning to address diabetic foot issues, such as classifying thermal pictures and predicting ulcer risk. By aggregating longitudinal sensor data, algorithms can be trained to identify nuanced risk trends.

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